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# Hospital Management System

def show_menu():
    print("\nHospital Management System")
    print("1. Add Patient")
    print("2. View Patients")
    print("3. Search Patient")
    print("4. Remove Patient")
    print("5. Exit")

def add_patient(patients, patient_set):
    name = input("Enter patient name: ")
    age = input("Enter patient age: ")
    disease = input("Enter disease: ")
    patient_id = len(patients) + 1
    patient_tuple = (patient_id, name, age, disease)
    patients.append(patient_tuple)
    patient_set.add(name)
    print("Patient added successfully!")

def view_patients(patients):
    if not patients:
        print("No patients found.")
        return
    print("\nList of Patients:")
    for p in patients:
        print(f"ID: {p[0]}, Name: {p[1]}, Age: {p[2]}, Disease: {p[3]}")

def search_patient(patients, name, index=0):
    if index >= len(patients):
        print("Patient not found.")
        return
    if patients[index][1].lower() == name.lower():
        print(f"Patient Found - ID: {patients[index][0]}, Name: {patients[index][1]}, Age: {patients[index][2]}, Disease: {patients[index][3]}")
        return
    search_patient(patients, name, index + 1)

def remove_patient(patients, patient_set):
    name = input("Enter patient name to remove: ")
    for p in patients:
        if p[1].lower() == name.lower():
            patients.remove(p)
            patient_set.discard(name)
            print("Patient removed successfully!")
            return
    print("Patient not found.")

def main():
    patients = []
    patient_set = set()
    while True:
        show_menu()
        choice = input("Enter your choice: ")
        if choice == '1':
            add_patient(patients, patient_set)
        elif choice == '2':
            view_patients(patients)
        elif choice == '3':
            name = input("Enter patient name to search: ")
            search_patient(patients, name)
        elif choice == '4':
            remove_patient(patients, patient_set)
        elif choice == '5':
            print("Exiting system...")
            break
        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()

...
Hospital Management System
1. Add Patient
2. View Patients

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3. Search Patient
4. Remove Patient
5. Exit
Enter your choice: 1
Enter patient name: Ankit Raj
Enter patient age: 19
Enter disease: hemorrhoids
Patient added successfully!
```

Hospital Management System

```
1. Add Patient
2. View Patients
3. Search Patient
4. Remove Patient
5. Exit
```

Enter your choice: